

CURRICULUM VITAE  
**YUHAO JIANG**

Post-doctoral Researcher, EPFL

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**CONTACT**

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**EDUCATIONAL EXPERIENCE**

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| <b>Arizona State University, Tempe, USA</b><br>Ph.D. in Mechanical Engineering<br><b>Advisor:</b> Prof. Daniel Aukes<br><b>Dissertation:</b> Design and Modeling of Soft Curved Reconfigurable Anisotropic Mechanisms | <i>Jan. 2019 - Aug. 2023</i>  |
| <b>University of Florida, Gainesville, USA</b><br>Master of Science in Mechanical Engineering                                                                                                                         | <i>Sept. 2015 - May 2017</i>  |
| <b>Donghua University, Shanghai, China</b><br>Bachelor of Engineering in Mechanical Engineering                                                                                                                       | <i>Sept. 2011 - Jun. 2015</i> |

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**PROFESSIONAL EXPERIENCE**

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| <b>EPFL, Lausanne, Switzerland</b><br>Post-doctoral Researcher, Reconfigurable Robotics Lab<br><b>Supervisor:</b> Prof. Jamie Paik | <i>Sept. 2023 - Present</i> |
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**PUBLICATIONS**

**Journal Publications**

- [1] **Y. Jiang**, F. Chen, J. Paik, and D. M. Aukes, "Vibration of Soft, Twisted Beams for Under-Actuated Quadrupedal Locomotion," IEEE/ASME Transactions on Mechatronics, Accepted, April 2026, [arxiv.org/abs/2507.02547](https://arxiv.org/abs/2507.02547).
- [2] X. Zheng, **Y. Jiang**, M. Mete, J. Li, I. Watanabe, T. Yamada, and J. Paik, "Metamaterial Robotics," Science Robotics, 10, eadx1519 (2025). [10.1126/scirobotics.adx1519](https://doi.org/10.1126/scirobotics.adx1519).
- [3] **Y. Jiang**, S. Asmar, Z. Wang, S. Demirtas, and J. Paik, "CPG-based Manipulation with Multi-Module Origami Robot Surface," IEEE Robotics and Automation Letters, March 2025, [10.1109/LRA.2025.3555381](https://doi.org/10.1109/LRA.2025.3555381).
- [4] **Y. Jiang**, F. Chen and D. M. Aukes, "Tunable Dynamic Walking via Soft Twisted Beam Vibration," IEEE Robotics and Automation Letters, vol. 8, no. 4, pp. 1967-1974, April 2023, [10.1109/LRA.2023.3244716](https://doi.org/10.1109/LRA.2023.3244716).
- [5] M. Sharifzadeh, **Y. Jiang**, A. Lafmejani, K. Nichols, and D. M. Aukes, "Maneuverable gait selection for a novel fish-inspired robot using a CMA-ES-assisted workflow," Bioinspiration & Biomimetics, vol. 16, no. 5, pp. 056017, August 2021, [10.1088/1748-3190/ac165d](https://doi.org/10.1088/1748-3190/ac165d).
- [6] M. Sharifzadeh, **Y. Jiang**, and D. M. Aukes, "Reconfigurable Curved Beams for Selectable Swimming Gaits in an Underwater Robot," IEEE Robotics and Automation Letters, vol. 6, no. 2, pp. 3437-3444, April 2021, [10.1109/LRA.2021.3063961](https://doi.org/10.1109/LRA.2021.3063961).

## Conference Publications

- [1] **Y. Jiang**, M. Sharifzadeh, and D. M. Aukes, “Shape Change Propagation Through Soft Curved Materials for Dynamically-Tuned Paddling Robots,” 2021 IEEE 4th International Conference on Soft Robotics (RoboSoft), 2021, pp. 230-237, 10.1109/RoboSoft51838.2021.9479208.
- [2] **Y. Jiang**, M. Sharifzadeh, and D. M. Aukes, “Reconfigurable Soft Flexure Hinges via Pinched Tubes,” 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020, pp. 8843-8850, 10.1109/IROS45743.2020.9341109.
- [3] D. Bailey, R. Moreno, S. Demirtas, Z. Wang, **Y. Jiang**, J. Paik, K. Stoy, and A. Faña. “Flexible and Foldable: Workspace Analysis and Object Manipulation Using a Soft, Interconnected, Origami-Inspired Actuator Array.” 2026 IEEE International Conference on Robotics and Automation (ICRA), 2026, Accepted, [arxiv.org/abs/2509.13998](https://arxiv.org/abs/2509.13998).
- [4] P. Bupe, **Y. Jiang**, J. Lin, T. Nguyen, M. Han, D. Aukes, C. Harnett, “Embedded Optical Waveguide Sensors for Dynamic Behavior Monitoring in Twisted-Beam Structures,” 2024 IEEE 7th International Conference on Soft Robotics (RoboSoft), San Diego, CA, USA, 2024, pp. 139-144, 10.1109/RoboSoft60065.2024.10521938.
- [5] M. Sharifzadeh, **Y. Jiang**, A. Lafmejani, D. M. Aukes, “Compensating for Material Deformation in Foldable Robots via Deep Learning – A Case Study,” 2022 IEEE International Conference on Robotics and Automation (ICRA), 2022, 10.1109/ICRA46639.2022.9811752.
- [6] M. Sharifzadeh, **Y. Jiang**, R. Khodambashi, D. M. Aukes, “Increasing the Life Span of Foldable Manipulators With Fabric.” Proceedings of the ASME 2020 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. Volume 10: 44th Mechanisms and Robotics Conference (MR). Virtual, Online. August 17–19, 2020. V010T10A087. ASME, 10.1115/DETC2020-22757.

## PATENTS

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- [1] “Pinched tubes for reconfigurable robots”, D. Aukes, M. Sharifzadeh, **Y. Jiang**, N. Gravish, M. Jiang - US Patent US20230127106A1
- [2] “Buckling beams for underwater and terrestrial autonomous vehicles”, D. Aukes, M. Sharifzadeh, **Y. Jiang** - US Patent US20230121727A1
- [3] “Mechanisms for steering robotic fish”, D. Aukes, M. Sharifzadeh, K. Nichols, **Y. Jiang** - US Patent US11124281B2
- [4] “Tunable Motion Using Flexible Twisted Beams”, D. Aukes, **Y. Jiang**, F. Chen - US Patent Application 20240391542

## SELECTED PROJECTS

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### **MOZART: Morphing Computerized mats with Embodied Sensing and Artificial Intelligence**

*Sep. 2023 - Present*

EPFL, Reconfigurable Robotics Lab

**Funded by:** EU Horizon Europe Research and Innovation Programme. Grant number: 101069536.

### **SCRAM: Soft Curved Reconfigurable Anisotropic Mechanisms**

*Jan. 2020 - May. 2023*

Arizona State University, IdeaLab

**Funded by:** NSF EFRI C3 SoRo. Grant number: 1935324.

### **FIRE: Fish-Inspired Robots for Extreme Environments**

*Jan.2019 - Jan. 2020*

Arizona State University, IdeaLab

**Funded by:** Salt River Project

## GRANT ACTIVITIES

| Proposal Title                                                   | Funding Source    | Amount | Role         | Status  | Period                |
|------------------------------------------------------------------|-------------------|--------|--------------|---------|-----------------------|
| COBRAS: Circular On-site Building with Robotic Autonomous Swarms | EU-EIC-PATHFINDER | EUR 4M | Co-Applicant | Granted | Nov. 2026 - Nov. 2029 |

## TEACHING AND STUDENT MENTORING

### Course Instructor

| Course Name                                                  | Affiliation | Period         |
|--------------------------------------------------------------|-------------|----------------|
| ME410: Mechanical Engineering Product Design and Development | ME, EPFL    | Fall 2023-2025 |
| ME420: Advanced Design for Sustainable Future                | ME, EPFL    | Fall 2024-2025 |

### Master's Thesis and Semester Project Advisor

| Name                        | Topic                                                                                                     | Program          | Period      |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|------------------|-------------|
| Goncalo Pais <sup>1</sup>   | Development of Energy Storage and Release Mechanisms for Rapid Dynamic Motions in Canfield Origami Robots | MS in Mechanical | Spring 2025 |
| Aurora Ruggeri <sup>1</sup> | Study on soft metamaterials for object sensing and geometry generation                                    | MS in Mechanical | Spring 2024 |
| Louis Flahault <sup>1</sup> | Kinematic study and design for spatial reconfigurable modular robotic platform                            | MS in Robotics   | Spring 2024 |
| Serge Asmar <sup>1</sup>    | Locomotion design and control using surface wave change generated by ori-pixel platform                   | MS in Robotics   | Spring 2024 |
| Nicolas Nouel <sup>2*</sup> | Programmable surface using bistable structure                                                             | MS in Robotics   | Spring 2024 |

<sup>1</sup> Master Semester Project <sup>2</sup> Master Thesis \* Co-advisor

## TALKS

### Conference Proceedings Talks

- [1] **IEEE-IROS 2025:** "CPG-based Manipulation with Multi-Module Origami Robot Surface"
- [2] **IEEE-RoboSoft 2023:** "Tunable Dynamic Walking via Soft Twisted Beam Vibration"
- [3] **IEEE-ICRA 2022:** "Compensating for Material Deformation in Foldable Robots Via Deep Learning – a Case Study", <https://youtu.be/AwS4vabv-JQ>
- [4] **IEEE-ICRA 2021:** "Reconfigurable Curved Beams for Selectable Swimming Gaits in an Underwater Robot", <https://youtu.be/EszTDc9slyw>
- [5] **IEEE-RoboSoft 2021:** "Shape Change Propagation Through Soft Curved Materials for Dynamically-Tuned Paddling Robots"
- [6] **IEEE-IROS 2020:** "Reconfigurable Soft Flexure Hinges via Pinched Tubes", <https://youtu.be/J5heXXD6mVo>

### Workshop Presentations

- [1] **IEEE-RoboSoft 2025:** "CPG-Based Motion Generator for Multi-Module Origami Robot"
- [2] **IEEE-RoboSoft 2023:** "Model Order Reduction for Vibrational Soft Twisted Beams Using Pseudo-rigid-body Modeling – A Case Study" (**awarded 2nd place for best talk**)
- [3] **IEEE-ICRA 2022:** "Modular Robots Using Soft Curved Reconfigurable Anisotropic Mechanisms"

## Seminar Talks

- [1] “Empowering Actuation of Soft Robotic Systems via Soft Curved Reconfigurable Anisotropic Mechanism”, hosted by Prof. Nick Gravish and Prof. Michael Tolley, UCSD, Feb. 2023.

## ACADEMIC SERVICE

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### Event Organization

- [1] **IEEE-IROS 2025 Workshop** : “S'MORE: Shape-Morphing Robotics via Embodied Sensing and Mechanisms”, <https://www.shapemorphingrobot.com/>
- [2] **2024 RRL Demo Day**: Full-day public event for research projects from RRL and ME-410 and ME-420 class,  
<https://www.paikslab.com/courses>
- [3] **2023 RRL Demo Day**: Full-day public event for research projects from RRL and ME-410 class,  
<https://www.paikslab.com/courses>
- [4] **Robosoft 2021 Workshop** : “Breaking the Mold: Challenging Current Paradigms in Soft Robotics”, <https://www.scrambots.com/robosoft-2021-workshop>

### Media Interview

- [1] **RTS Education and Scientific Program**: feature in “A guide to the future: Swiss Federal Institute of Technology 02”, <https://www.youtube.com/watch?v=9yoNLg5Qho0&t=560s>

### Editor

IEEE-IROS 2026: Associate Editor

### Journal Reviewer

The International Journal of Robotics Research (IJRR)  
IEEE Transactions on Robotics (T-RO)  
IEEE/ASME Transactions on Mechatronics (T-Mech)  
IEEE Robotics and Automation Letters (RA-L)  
Soft Robotics (SoRo)  
Journal of Field Robotics (JFR)  
ASME Journal of Mechanisms and Robotics (JMR)

### Conference Reviewer

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)  
International Conference on Robotics and Automation (ICRA)  
International Conference on Soft Robotics (Robosoft)  
ACM Symposium on Computational Fabrication (SCF)

### Grant Reviewer

Swiss National AI Institute (Seed Grant Program)

### Demos and Expositions

- [1] Swiss Robotics Day (2023 - 2025)